Fiscal Year:	FY 2019	**Task Last Updated:**	FY 01/25/2022
PI Name:	Beckage, Michael B.S.		
Project Title:	Crew Worn Acceleration Recorder for Spaceflight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	mike.beckage@dtsweb.com	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	562-493-0158
Organization Name:	Diversified Technical Systems, Inc.		
PI Address 1:	1720 Apollo Court		
PI Address 2:			
PI Web Page:			
City:	Seal Beach	**State:**	CA
Zip Code:	90740-5617	**Congressional District:**	48
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	SBIR Phase II
Start Date:	08/14/2019	**End Date:**	08/13/2021
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC19C0119		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	As early as the Gemini missions, NASA has monitored physiological parameters during spaceflight for the collection of real time and recorded data during critical missions. This has helped NASA better understand human reactions to space by monitoring astronauts' temperature, respiration, and cardiac activity. The proposed Dynamic Kinematic Recorder (DKR) is designed to measure the biomechanical vibration, impact and motion of astronauts' head, neck, body, and helmet. Astronauts can be subjected to excessive vibration and shock during launch, re-entry, or abort scenarios. Biomechanical measurements in these conditions have unique challenges. Safety of flight is essential and the proposed DKR is required to be an ultra-low power system that minimizes size and weight and runs autonomously. Diversified Technical Systems, Inc. (DTS) has unique experience developing similar systems over the past 27 years, including work with NASA measuring crew biomechanics. DTS has developed systems to measure the kinematic motion of rodeo riders, aerobatic pilots, ejection seat manikins, roller coaster riders, soldiers in combat, and most recently football		

	players through a research project funded by the National Football League (NFL). All DTS systems are designed in accordance with the Society of Automotive Engineers (SAE) J211 recommendations. Recent technology advances in Micro Electro-Mechanical System (MEMS) sensing and microprocessor technology now make it possible to deliver a relatively low cost, small, light, and autonomous system to measure crew biomechanics. Unique to this proposal is a technical breakthrough in angular sensing technology that reduces size and power by orders of magnitude. DTS proposes to deliver a revolutionary prototype 6 degrees of freedom (DOF) Dynamic Kinematic Recorder that will have greater than 10-fold reduction in size and mass compared to existing systems. This autonomous system would be small, ultra-low power, and weigh less than 20 grams.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Potential Non-NASA Applications: The Dynamic Kinematic Recorder (DKR) has wide application in other industries. First and foremost is the measurement of head motion to help understand traumatic brain injury in sports and soldiers in blast and field environments. Other applications include high value asset monitoring during shipping. A DKR could be attached to the asset to monitor shock and vibration levels during transit.
Task Progress:	New project for FY2019. Reporting not required for this SBIR Phase II project. (Ed. note: added to Task Book on 01/20/2022)
Bibliography Type:	Description: (Last Updated:)